

RUIWEN ZHOU

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EDUCATION

Ph.D. student in Computer Science, National University of Singapore	Aug 2025 – Present
M.Eng. in Computer Science, Shanghai Jiao Tong University	Sep 2022 – Mar 2025
B.Eng. in Information Engineering, Shanghai Jiao Tong University	Sep 2018 – Jun 2022

RESEARCH INTERESTS

Core Areas: Large Language Models (LLMs), AI Agents, Reinforcement Learning (RL).

Research Focus:

- AI agents (for coding, research, etc.) and multi-agent systems.
- Reinforcement learning and retrieval augmentation for (agentic) LLMs.
- LLM reasoning within complex context and social environments.

SKILLS

- **Languages:** Python, C++, L^AT_EX.
- **Frameworks:** PyTorch, Transformers, vLLM, VeRL, TRL.

EXPERIENCE

Ph.D. Student, Supervised by: Prof. Min-Yen Kan & Prof. Soujanya Poria WING Lab (National University of Singapore)	Aug 2025 – Present <i>Singapore</i>
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- **Research Focus:** LLM-based Agents, Multi-Agent Systems, RL for LLM Reasoning.
- **Project: Trust Estimation and Utilization in LLM-Based Multi-Agent Systems**
 - Investigated LLMs' sycophancy and inadequate ability to evaluate peer reliability in multi-agent systems, and formulated the learning problem of *history-aware reference* as an alternative solution, which introduces the historical interactions of peers as additional input and shifts the task from evaluating peer reasoning quality to estimating peer reliability based on interaction history.
 - Developed *Epistemic Context Learning* (ECL), a reasoning framework that conditions predictions on explicitly-built peer profiles from history, and further proposed to optimize ECL by RL using auxiliary rewards.
 - Conducted extensive experiments to show that ECL enables Qwen 3-4B to outperform a history-agnostic baseline 8x its size (Qwen 3-30B) and boosts frontier models to near-perfect (100%) performance, and did additional analysis that provides insights on how historical trust is learned and utilized.
- **Under Review.**
- **Project: Measuring and Mitigating Rapport Bias in Multi-Agent Interactions**
 - Investigated the vulnerability of LLMs to social rapport bias and peer pressure, which often compromises factual accuracy in group settings.
 - Contributed to construction of KAIROS benchmark and evaluated GRPO with outcome rewards as a mitigation strategy, validating that RL-tuned LLMs can improve over base models in resisting social conformity.
- **Accepted at ICLR 2026.**

- **Research Focus:** LLM Reasoning, LLM Evaluation
- **Project: LLM Reasoning under Natural Language Rule Guidance**
 - Addressed the limitation of existing benchmarks that focus on simple first-order logic, failing to evaluate LLMs' adherence to complex, real-world constraints (e.g., tax/legal rules).
 - Led the creation of RuleArena, a benchmark requiring long-context understanding and multi-hop reasoning based on real-world applications (airlines, NBA, and tax). Developed rigorous evaluation metrics to decouple rule retrieval errors from rule reasoning errors.
 - Conducted comprehensive evaluation of mainstream LLMs (GPT-4o, Claude-3.5, etc.), revealing critical gaps in handling high-order logic constraints compared to first-order logic tasks.
 - **Published at ACL 2025 Main Conference.**
- **Project: Contamination-Free LLM Evaluation**
 - Mitigated the critical "Data Contamination" issue where static benchmarks leak into training data, rendering evaluations unreliable.
 - Co-developed a fully automated pipeline that constructs evaluation samples using dynamic, time-sensitive knowledge (from Wikidata/Wikipedia updates post-training cutoff).
 - **Published at ACL 2025 Main Conference — Selected as SAC Highlight.**

- **Project: Retrieval-Augmented LLM Agents**
 - Addressed the issue of "plausible but wrong" and noisy in-context example in trajectory-level retrieval for long-horizon agentic tasks.
 - Developed TRAD, a novel agent framework combining Step-wise Thought Retrieval (using CoT as search queries and keys) and Aligned Decision (for concentrating on key trajectory segments) mechanisms.
 - Achieved SOTA performances on ALFWorld and Mind2Web. Successfully deployed in enterprise RPA scenarios at China Pacific Insurance Company (CPIC), improving task success rate from 65.0% to 92.5%.
 - **Published at SIGIR 2024.**
- **Project: Risk-Sensitive Distributional RL**
 - Identified a fundamental "biased optimization" flaw in existing Risk-Sensitive RL (RSRL) where standard Bellman operators fail to converge for general distortion risk measures.
 - Proposed Trajectory Q-Learning (TQL), a theoretically rigorous algorithm modeling history-dependent return distributions to guarantee convergence to the optimal risk-sensitive policy, the first general solution for optimizing diverse risk measures (e.g., CVaR) with provable optimality.
 - **Work done during internship at MSRA — Published at DAI 2025.**
- **Project: Retrieval-Augmented Graph Representation Learning**
 - Tackled the limitation of independent instance modeling in tabular data prediction tasks.
 - Co-designed a Hypergraph Neural Network utilizing retrieval-based neighborhood propagation to explicitly model high-order cross-row feature and label interactions.
 - Achieved SOTA results on CTR prediction and Top-N recommendation benchmarks.
 - **Work done during internship at AWS — Published at NeurIPS 2022.**

PUBLICATIONS

- [1] **Ruiwen Zhou**, Maojia Song, Xiaobao Wu, Sitao Cheng, Xunjian Yin, Yuxi Xie, Zhuoqun Hao, Wenye Hua, Liangming Pan, Soujanya Poria, Min-Yen Kan. [Epistemic Context Learning: Building Trust the Right Way in LLM-Based Multi-Agent Systems](#). *arXiv preprint arXiv:2601.21742*.
- [2] Shengtao Zhang, Jiaqian Wang, **Ruiwen Zhou**, Junwei Liao, Yuchen Feng, Weinan Zhang, Ying Wen, Zhiyu Li, Feiyu Xiong, Yutao Qi, Bo Tang, Muning Wen. [MemRL: Self-Evolving Agents via Runtime Reinforcement Learning on Episodic Memory](#). *arXiv preprint arXiv:2601.03192*.
- [3] Sitao Cheng, Xunjian Yin, **Ruiwen Zhou**, Yuxuan Li, Xinyi Wang, Liangming Pan, William Yang Wang, Victor Zhong. [From Atomic to Composite: Reinforcement Learning Enables Generalization in Complementary Reasoning](#). *arXiv preprint arXiv:2512.01970*.
- [4] Maojia Song, Tej D. Pala, **Ruiwen Zhou**, Weisheng Jin, Amir Zadeh, Chuan Li, Dorien Herremans, Soujanya Poria. [Measuring and Mitigating Rapport Bias of LLMs under Multi-Agent Social Interactions](#). *ICLR 2026*.
- [5] **Ruiwen Zhou**, Wenye Hua, Liangming Pan, Sitao Cheng, Xiaobao Wu, En Yu, William Yang Wang. [RuleArena: A Benchmark for Rule-Guided Reasoning with LLMs in Real-World Scenarios](#). *ACL 2025 (Main)*.
- [6] Xiaobao Wu, Liangming Pan, Yuxi Xie, **Ruiwen Zhou**, Shuai Zhao, Yubo Ma, Mingzhe Du, Rui Mao, Anh Tuan Luu, William Yang Wang. [AntiLeak-Bench: Preventing Data Contamination by Automatically Constructing Benchmarks with Updated Real-World Knowledge](#). *ACL 2025 (Oral & SAC Highlight)*.
- [7] **Ruiwen Zhou**, Yingxuan Yang, Muning Wen, Ying Wen, Wenhao Wang, Chunling Xi, Guoqiang Xu, Yong Yu, Weinan Zhang. [TRAD: Enhancing LLM Agents with Step-Wise Thought Retrieval and Aligned Decision](#). *SIGIR 2024 (Oral)*.
- [8] **Ruiwen Zhou**, Minghuan Liu, Kan Ren, Xufang Luo, Weinan Zhang, Dongsheng Li. [Is Risk-Sensitive Reinforcement Learning Properly Resolved?](#) *arXiv preprint arXiv:2307.00547*.
- [9] Kounianhua Du, Weinan Zhang, **Ruiwen Zhou**, Yangkun Wang, Xilong Zhao, Jiarui Jin, Quan Gan, Zheng Zhang, David Wipf. [Learning Enhanced Representations for Tabular Data via Neighborhood Propagation](#). *NeurIPS 2022*.

SELECTED AWARDS

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| • NUS Graduate Research Scholarship | 2025 |
| • Huatai Securities Fellowship | 2024 |
| • First-Class Excellence Scholarship | 2022, 2024 |
| • Zhiyuan Honors Scholarship (Top 5%) | 2019, 2020, 2021 |
| • China National Scholarship (Rank 1/144) | 2020 |
| • A-Class Excellence Scholarship (Rank 1/144) | 2020 |